

TATH SIKDAR

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Education

McMaster University

B.Eng in Mechatronics Engineering Co-op

Sep 2021 - Nov 2026

Hamilton, ON

Work Experience

Nestlé Canada

AI Full Stack Developer

Apr 2024 - Dec 2025

Toronto, ON

- Centralized **14** competitor data sources into a **PostgreSQL** warehouse by building **Python ETL data pipelines** with custom **data cleaning** rules, cutting duplicate listings **23%** and ending double counted price moves.
- Deployed a production-ready **machine learning classification** model over 40,000+ SKUs at **94%** accuracy against a **71%** majority-class baseline, by training **scikit-learn** on **TF-IDF** features engineered in **Pandas**.
- Replaced a manual weekly **Excel** reporting process with self-serve **Power BI** and **Tableau** dashboards on a star-schema **SQL** warehouse, returning roughly **6** analyst hours a week across **500+** enterprise users.
- Diagnosed a **9%** degradation in categorization precision caused by an unannounced supplier feed change, utilizing **model drift monitoring** and **K-means clustering** of misclassified SKUs to prevent downstream data disruption.

Chase Auto Parts

Software Developer Co-op

Sep 2023 - Apr 2024

Mississauga, ON

- Cut part-identification errors **57%** against the incumbent lookup flow through **feature engineering** on an existing live transcription feed, scoring precision and recall on a held-out set of 4K annotated conversations.
- Ruled out **SARIMA** and Prophet for time series demand **forecasting** after benchmarking both in **Python** and **R**; only a **gradient boosting** model beat the incumbent reorder rule, cutting projected stockouts **18%** in backtest.
- Traced most of the cloud bill to three query patterns through **exploratory data analysis** and **data wrangling** of raw **Azure** telemetry; rewriting those sync jobs with platform engineering cut redundant calls **60%**.
- Engineered a fault-tolerant, cross-platform (**Linux/Windows**) nightly forecast pipeline by implementing idempotent writes and automated **data quality** checks to eliminate data gaps and prevent inventory double-counting during reruns.

Fanique Group

Application Developer Co-op

May 2023 - Sep 2023

Toronto, ON

- Increased user session duration **25%** by identifying the winning onboarding flow from a 6-variant **A/B test**, sizing each cohort for **80%** power and grounding the readout in applied **statistics** and **statistical modeling**.
- Isolated a **30%** Day-7 retention gap between web and mobile cohorts by instrumenting a typed cross-platform event schema, then running **cohort** and **survival analysis** in **Python** to scope the product fix the team shipped.

Personal Projects

🐙 Conformal Cashflow - Liquidity Forecasting | *Python, PyTorch, Conformal Prediction, SQL, Docker*

- Achieved **90% conformal coverage (PICP)** on liquidity forecasts by building a two-stage hurdle model, a classification gate feeding a quantile head, to put calibrated intervals around every prediction rather than a point estimate.
- Scaled training to **1.1M** transaction records by engineering per-account normalization, cyclical time encodings, and **FFT** spectral features in **NumPy** and **SciPy**, holding sharpness across heterogeneous accounts.

🐙 RouteWise - Fuel Economy Prediction | *Python, LightGBM, Random Forest, Optuna, SHAP, SQLite*

- Improved fuel-economy R^2 from **0.42** to **0.84** by cascading two **regression** models to separate congestion effects from vehicle physics, using Vehicle Specific Power as the bridging feature across 260+ vehicles.
- Tuned the **LightGBM** and **Random Forest** ensemble with **Optuna** Bayesian search, then read **SHAP** attributions to confirm the model leaned on road grade and speed variance rather than dataset collection artifacts.

Skills

Languages: Python, SQL, R, Scala, Julia, Java, C, C++

Libraries: scikit-learn, PyTorch, LightGBM, PySpark, Pandas, NumPy, SciPy, Optuna, SHAP, Matplotlib

Methods: Random Forest, A/B testing, experimental design, statistical modeling, conformal prediction, cohort analysis, survival analysis, statistics, regression, classification, clustering, forecasting, SARIMA

Platforms: Power BI, PostgreSQL, Databricks, SQLite, Tableau, Excel, MATLAB, Azure, AWS, GCP, Docker